

01.1 • Probability through an AI-assisted ticket-triage problem

Data Analytics for AI • Executive M.Tech • IIT Jammu

Module 01 • Fundamentals of Probability and Random Signal Theory

01.1 • Saturday, 3 October 2026 • 10:30 a.m.–12:10 p.m. IST

Lecture materials

- **Notes:** PDF
- **Deck:** Worked student deck (PDF)
- **Interactive lab:** Open the probability lab

Supplementary material

- **Notes:** Rigorous supplement (PDF)
- **Deck:** Probability before Kolmogorov (PDF)
- **Deck:** Bertrand's paradox (PDF)
- **Interactive demonstration:** Sampling maps and chord length

Purpose and pace

We begin with a six-ticket routing audit and use it to motivate the mathematical objects needed to describe events and assign probabilities. The sequence is:

1. Compare reviewer and model classifications, including a ticket with two required destinations.
2. Compare historical interpretations through the audit's agreement fraction.
3. Pose Bertrand's random-chord question and note its different answers under different sampling procedures.
4. Introduce Kolmogorov's set-based probability space, then specify uniform selection from the six tickets and define events as sets.
5. Introduce the triplet's requirements, a random variable and its law.
6. Calculate a continuous-time probability by integration.
7. Practise the same definitions in weighted audits, finance, engineering and training.

The six ticket IDs connect the main argument; the practice problems vary the context. The meeting duration does not set a required pace; stop at a checkpoint and continue at the next meeting when needed.

1. A ticket-routing audit

An IT service desk is checking whether an LLM routes incoming tickets to the appropriate department. The pilot uses three reference outcomes:

- **Billing** when Billing is required;
- **Account Support** when Account Support is required;
- **Other** when neither Billing nor Account Support is required.

“Other” is the complement category for this two-destination question: the reviewer finds that neither Billing nor Account Support is required. Another team may handle such a ticket; “Other” does not mean that the ticket was unreviewed or its status is unknown. A ticket may require both Billing and Account Support, so the reference outcome can contain both labels.

The table is a hypothetical audit. Reviewer labels are operational reference decisions; they are not asserted to be error-free ground truth. For t_1 , the LLM reports scores $s_B(x) = 0.55$ and $s_A(x) = 0.35$ for Billing and Account Support. These are model scores; calibration as probabilities is not assumed. To obtain the hard model classification shown in the table, apply a threshold of 0.50 separately to each destination. The other rows show the resulting model classifications.

Ticket	Request summary	Reviewer reference	Model classification
t_1	"I was charged twice, and now I cannot log in to my account."	Billing and Account Support	Billing
t_2	A duplicate charge appears on the monthly invoice	Billing	Billing
t_3	Cannot log in after changing the phone used for MFA	Account Support	Account Support
t_4	A delivery-status question unrelated to billing or account access	Other	Account Support
t_5	A subscription renewal was charged twice	Billing	Billing
t_6	Account locked after a password reset	Account Support	Account Support

The model and reviewer agree exactly on t_2, t_3, t_5, t_6 . They disagree on t_1 , where the model omits Account Support, and t_4 , where the model assigns Account Support but the reviewer assigns Other. Relative to the reviewer protocol, the Account Support decision on t_1 is a false negative and that on t_4 is a false positive. Thus the observed agreement fraction is 4/6.

Ask:

What does 4/6 describe? Under what experiment would it be a probability, and what additional assumptions would connect it to future tickets?

The fraction is an observed summary of these six records. It becomes a probability for selecting uniformly from these six ticket IDs once that selection experiment is specified. It does not, by itself, describe future service-desk traffic.

Checkpoint 1. Identify the reference outcome for t_1 and explain why “Other” applies to t_4 but not to t_1 .

2. Historical interpretations of probability

Probability has been interpreted in several ways. The accounts below answer different questions about what the number 4/6 might mean. None follows just from seeing six rows in a table.

Classical equal-likelihood

The classical rule counts favourable outcomes among equally likely cases. If one ticket ID is selected uniformly from the six IDs, four selections produce agreement, so

$$P(\text{agreement}) = \frac{4}{6} = \frac{2}{3}.$$

This calculation is valid for that selection experiment. It does not imply that future tickets are equally likely to be any of these six records, or that the six records represent future traffic. A finite list defines possible cases; the uniform selection rule supplies the equal weights.

The partition chosen for counting also matters. Group the tickets by the pair (reviewer outcome, model classification):

Reviewer outcome	Model classification	Ticket IDs	Exact match?
Billing and Account Support	Billing	t_1	No
Billing only	Billing	t_2, t_5	Yes
Account Support only	Account Support	t_3, t_6	Yes
Other	Account Support	t_4	No

There are four occupied pair types in this audit. If an experiment selected one of these four types uniformly, two of the four would be exact matches, giving 2/4. If it selected one of the six ticket IDs uniformly, the pair types would instead have probabilities 1/6, 2/6, 2/6, 1/6, giving 4/6. Thresholding two destination scores separately permits four model outputs: neither, Billing only, Account Support only, or both. Thus the full product contains $4 \times 4 = 16$ reviewer/model pairs, of which four occur in this audit. Uniform selection among all 16 pairs would be a third experiment; its four matching pairs would give agreement probability $4/16 = 1/4$. Equal weighting of pair types does not follow from grouping tickets that way. Counting requires a stated set of cases and a reason to treat those cases as equally likely.

Frequency

The observed agreement proportion is

$$\widehat{p}_6 = \frac{1}{6} \sum_{i=1}^6 I_i = \frac{4}{6},$$

where $I_i = 1$ when the model classification exactly matches the reviewer’s reference outcome for ticket t_i , and $I_i = 0$ otherwise. A frequency interpretation connects a probability to the limiting relative frequency in a specified repeatable process. This audit records one finite set of six tickets; it does not establish that limit or show that future ticket mix and model performance will remain the same.

Repeatedly sampling the LLM for the same text under randomized decoding would define another process. Its output frequency would describe generated classifications, not necessarily agreement with reviewer decisions. The process and event must be stated before interpreting a frequency.

Subjective probability and Bayesian updating

A subjective interpretation treats probability as a person’s degree of belief given available information. An analyst might assess the chance that the model agrees with the review protocol on a future ticket, using this audit and other stated information. The six rows do not determine that assessment uniquely. Coherence constrains probability assignments; subjective does not mean arbitrary. Countable additivity is an additional condition in the Kolmogorov model and does not follow from every finite coherence argument.

Bayesian updating describes how an assessment changes when evidence is observed. If H is the proposition that the next ticket’s model classification will agree with its reference outcome and D is additional evidence, Bayes’ rule gives

$$P(H \mid D) = \frac{P(D \mid H)P(H)}{P(D)}, \quad P(D) > 0.$$

The formula does not supply the prior or likelihood; those come from a model and stated information. Conditional probability and Bayesian updating will be developed later.

Logical or evidential probability

A logical or evidential interpretation aims to quantify how strongly stated evidence supports a proposition. Here the evidence could include the six audit rows, the review protocol and assumptions about the future ticket population. A numerical answer requires a rule connecting that evidence and those assumptions to a degree of support. If the rule invokes indifference, it must specify which cases are treated equally and why. Counting ticket IDs and counting occupied label-pair types give different answers because they describe different selection rules. This identifies a problem with an unspecified indifference rule; it does not refute every logical account.

What the interpretations leave for a probability model

Classical probability emphasizes equal likelihood, frequency emphasizes repetition, subjective probability represents an assessment given information, and logical accounts specify evidential support. A propensity account would treat the configured randomized generation process as the chance-producing mechanism. These accounts differ in interpretation. A mathematical calculation also needs the possible outcomes, the event of interest and a probability assignment. The next example shows why the sampling procedure is part of that specification.

Checkpoint 2. For each interpretation, state what additional assumption or information is needed to use it for a claim about future tickets.

3. A second question: Bertrand's random chord

Draw an equilateral triangle inside a circle. Consider the question:

If a chord is selected at random from the circle, what is the probability that its length exceeds the side length of the triangle?

The phrase "selected at random" does not specify how the chord is generated. Bertrand described different sampling procedures for chords, which give $1/3$, $1/2$ and $1/4$ for the same length comparison. These values follow from different probability distributions induced by the respective sampling procedures; the problem statement alone does not say which procedure to use. John D. Norton's University of Pittsburgh page presents the three sampling arguments and their values.

The procedures are: choose two independent uniform endpoints ($1/3$); choose a uniform radial distance and an independent uniform direction ($1/2$); or choose a midpoint uniformly in disk area ($1/4$). These are different models. We will not choose among them or derive the answers in this section. The example identifies information a probability question can leave unspecified: the outcomes and, especially, the sampling rule that determines their probabilities.

Kolmogorov's set-based framework

Kolmogorov's axiomatic framework gives probability a mathematical foundation in terms of sets and a measure. Its three objects are written

$(\Omega, \mathcal{F}, P)$.

Here Ω is the sample space of possible outcomes, $\mathcal{F}$ is the collection of events to which probabilities are assigned, and $P: \mathcal{F} \rightarrow [0, 1]$ is the probability measure. The event collection has closure requirements, and the measure obeys nonnegativity, normalization and countable additivity. We will define and apply each object using the ticket audit. The supplement returns to Bertrand's question and specifies the sampling models in full.

4. Define the experiment and its events

To calculate a probability for the audit, specify an experiment:

Select one of the six ticket IDs uniformly at random.

The ticket records, reviewer decisions and model classifications are fixed. The selected ticket ID is the outcome that varies. Thus the sample space is

$$\Omega = \{t_1, t_2, t_3, t_4, t_5, t_6\}.$$

Define events from the reviewer's reference classification:

$$B = \{t_1, t_2, t_5\}, \quad A = \{t_1, t_3, t_6\},$$

where B is the event "Billing is required" and A is the event "Account Support is required." The event that neither is required is

$$O = (B \cup A)^c = \{t_4\}.$$

In this model, O is precisely the "Other" reference outcome. It is disjoint from B and A . The ticket t_1 belongs to both B and A , because both destinations are required:

$$B \cap A = \{t_1\}, \quad B \cup A = \{t_1, t_2, t_3, t_5, t_6\}.$$

For sets $E, F \subseteq \Omega$:

$$E \cap F = \{t \in \Omega: t \in E \text{ and } t \in F\},$$

$$E \cup F = \{t \in \Omega: t \in E \text{ or } t \in F\},$$

$$E^c = \Omega \setminus E.$$

The "or" in a union is inclusive. The outcome t_1 is counted in both marginal events, so to count tickets requiring at least one named destination we use the disjoint pieces $B \setminus A$, $A \setminus B$ and $B \cap A$:

$$B \cup A = (B \setminus A) \cup (A \setminus B) \cup (B \cap A).$$

Under uniform selection, their probabilities are $2/6$, $2/6$ and $1/6$. Therefore

$$P(B \cup A) = \frac{2}{6} + \frac{2}{6} + \frac{1}{6} = \frac{5}{6}.$$

Equivalently,

$$P(B \cup A) = P(B) + P(A) - P(B \cap A) = \frac{3}{6} + \frac{3}{6} - \frac{1}{6} = \frac{5}{6}.$$

The complement is O , so $P(O) = 1/6$. The reviewer outcomes form four possible destination sets: both, Billing only, Account Support only, or neither. "Other" refers only to the last of these in this example.

The exact-agreement event is $C = \{t_2, t_3, t_5, t_6\}$, so $P(C) = 4/6$ under this same uniform-ID experiment. We will also represent agreement numerically.

Checkpoint 3. Give the ticket IDs in B , A , $B \cap A$, $B \cup A$ and O . Explain why the probability of the union is not the sum of the two marginal probabilities.

5. The probability space and its requirements

The experiment is represented by the probability triplet

$$(\Omega, \mathcal{F}, P).$$

Object	Ticket-audit model	Meaning
Ω	$\{t_1, t_2, t_3, t_4, t_5, t_6\}$	Possible outcomes: ticket IDs that may be selected
$\mathcal{F}$	2^Ω	Events: all subsets of the six ticket IDs
P	$P(E) = E /6 \text{ for } E \subseteq \Omega$	Probability measure under uniform selection

An outcome such as t_2 is an element of Ω . An event such as B is a subset of Ω . The measure P takes an event as input and returns a number in $[0, 1]$. Here the uniform rule assigns probability $1/6$ to each singleton outcome.

The event collection $\mathcal{F}$ must be a **sigma-algebra** on Ω . Its requirements are:

1. **Contains the sample space:** $\Omega \in \mathcal{F}$.
2. **Closed under complements:** if $E \in \mathcal{F}$, then $E^c = \Omega \setminus E \in \mathcal{F}$.
3. **Closed under countable unions:** if $E_1, E_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} E_i \in \mathcal{F}$.

These conditions ensure that “not E ” and “at least one of $E_1, E_2, \dots$ ” are events whenever the component sets are events. Since this example has a finite sample space, taking $\mathcal{F} = 2^\Omega$, the power set, satisfies all three requirements.

The probability measure satisfies Kolmogorov’s axioms:

1. **Nonnegativity:** $P(E) \geq 0$ for each $E \in \mathcal{F}$.
2. **Normalization:** $P(\Omega) = 1$.
3. **Countable additivity:** if $E_1, E_2, \dots$ are pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i).$$

For the six singleton events, additivity gives $P(\Omega) = 6(1/6) = 1$. Also B and B^c are disjoint and exhaustive, and $P(B) + P(B^c) = 3/6 + 3/6 = 1$. Closure requirements apply to $\mathcal{F}$; the three axioms apply to P .

A probability measure is a set function from events to the unit interval. The event B is a set of tickets; $P(B) = 1/2$ is a number. This keeps the sample point, event and probability value distinct.

A smaller event collection

Suppose a dashboard records only whether the selected ticket requires Billing. It distinguishes $B = \{t_1, t_2, t_5\}$ from $B^c = \{t_3, t_4, t_6\}$, but does not identify individual tickets. The events available in this description are

$$\mathcal{G} = \{\emptyset, B, B^c, \Omega\}.$$

This is a sigma-algebra: complements exchange the pairs, and any union of its members is again one of the four members. Restrict the uniform-ID measure to $\mathcal{G}$; the four probabilities are 0, 1/2, 1/2, 1 respectively. The singleton $\{t_1\}$ is a subset of Ω but is not an event in $\mathcal{G}$. The full audit model still uses $\mathcal{F} = 2^\Omega$; the smaller model illustrates why the event collection must be specified.

Checkpoint 4. State the sample space, event collection and probability measure. Verify nonnegativity, normalization and additivity for two disjoint events from the table.

A continuous triplet on the unit interval

Suppose a simulation selects a number uniformly from the unit interval. What is the probability that the selected value lies between $1/4$ and $3/4$? Uniformity means that an interval’s probability is its length, so the answer should be $3/4 - 1/4 = 1/2$. To make this a probability model, we must specify more than intervals: “outside this interval” and “in at least one of these intervals” must also be events.

Take $\Omega_u = [0, 1]$. A sample point $u \in [0, 1]$ is a number; an event is a suitable **set of numbers**. We use the Borel σ -algebra

$$\mathcal{F}_u = \mathcal{B}([0, 1]) = \sigma\left(\{(a, b) \cap [0, 1] : a, b \in \mathbb{R}, a < b\}\right).$$

Here $\sigma(\mathcal{A})$ means the smallest σ -algebra containing every member of a family $\mathcal{A}$. Equivalently, it is the intersection of all σ -algebras on $[0, 1]$ that contain that family. Such σ -algebras exist because the power set $2^{[0, 1]}$ is one of them. The intervals intersected with $[0, 1]$ generate all the relatively open sets of the unit interval.

A **Borel set** is one member of $\mathcal{B}([0, 1])$. The **Borel σ -algebra** is the collection of all those members. For example,

$$[1/4, 3/4] \in \mathcal{B}([0, 1]), \quad [1/4, 3/4] \subseteq [0, 1].$$

The first statement says it is a measurable event; the second says its elements are possible outcomes.

What the Borel σ -algebra contains

It contains $\emptyset$, $[0, 1]$, every relatively open or closed subset, every interval within $[0, 1]$ (including half-open intervals), and every singleton $\{u\}$. It is closed under complements **relative to** $[0, 1]$, countable unions and countable intersections. The intersection property follows from De Morgan's law:

$$\bigcap_{n=1}^{\infty} E_n = [0, 1] \setminus \bigcup_{n=1}^{\infty} ([0, 1] \setminus E_n).$$

For example, each set $[0, 1] \cap (u - 1/n, u + 1/n)$ is relatively open, and

$$\{u\} = \bigcap_{n=1}^{\infty} ([0, 1] \cap (u - 1/n, u + 1/n)).$$

Thus singletons are Borel. Every countable set of points in $[0, 1]$ is then Borel, being a countable union of singletons. In particular, $\mathbb{Q} \cap [0, 1]$ is Borel, as is its complement, the irrational points in $[0, 1]$.

These examples do not exhaust the collection: The collection includes sets formed through repeated countable unions, intersections and complements, beyond the examples listed. Closure under **countable** operations does not imply closure under arbitrary uncountable unions. The Borel σ -algebra is smaller than $2^{[0,1]}$; not every subset is Borel. The supplement's Section 5 constructs a set that is not Lebesgue measurable. Since every Borel set is Lebesgue measurable, that set is not Borel either.

Assigning probabilities and checking the triplet

Let λ be Lebesgue measure, the standard measure that assigns each interval its length, and restrict it to Borel subsets of $[0, 1]$:

$$P_u(E) = \lambda(E), \quad E \in \mathcal{B}([0, 1]).$$

No scaling is needed because $\lambda([0, 1]) = 1$. Our triplet is

$$([0, 1], \mathcal{B}([0, 1]), P_u).$$

To verify it, check the event collection and the measure separately.

- **Event collection.** Every σ -algebra in the defining intersection contains $[0, 1]$ and is closed under relative complements and countable unions. Their intersection inherits these properties, so $\mathcal{B}([0, 1])$ is a σ -algebra.
- **Nonnegativity.** Lebesgue measure is nonnegative, so $P_u(E) \geq 0$ for each Borel event E .
- **Normalization.** $P_u([0, 1]) = \lambda([0, 1]) = 1$.
- **Countable additivity.** For pairwise disjoint Borel events $E_1, E_2, \dots$, the countable additivity of Lebesgue measure gives

$$P_u\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P_u(E_n).$$

Here we use the standard theorem that Lebesgue measure exists and is countably additive on Borel sets. Checking a few intervals alone would not establish that theorem. Restricting this measure to the unit interval verifies all the probability axioms, and $P_u(E) \leq 1$ follows from $E \subseteq [0, 1]$.

Worked events in the triplet

Borel event E	Probability $P_u(E)$	Reason
$[1/4, 3/4]$	1/2	Interval length
$[0, 1/4) \cup (3/4, 1]$	1/2	Complement of the preceding event
$\{1/2\}$	0	A singleton has zero length
$\mathbb{Q} \cap [0, 1]$	0	Countably many disjoint singletons of measure zero
$[0, 1] \setminus \mathbb{Q}$	1	Complement of the rational points

The rational points are dense in $[0, 1]$, but still have probability zero in this model. The measure is length, not the number of points or how closely they are spaced.

For an explicit countable-additivity calculation, let

$$I_n = (2^{-(n+1)}, 2^{-n}], \quad n = 0, 1, 2, \dots$$

These Borel intervals are pairwise disjoint and have union $(0, 1]$. Hence

$$P_u((0, 1]) = \sum_{n=0}^{\infty} 2^{-(n+1)} = 1.$$

Adding the zero-probability singleton $\{0\}$ gives $P_u([0, 1]) = 1$. This example illustrates countable additivity. It does not license an uncountable sum of singleton probabilities.

6. A random variable for the same audit

The reviewer event B can be represented by a numerical function. Define the Billing indicator

$$X_B: \Omega \rightarrow \mathbb{R}, \quad X_B(t) = \begin{cases} 1, & t \in B, \\ 0, & t \notin B. \end{cases}$$

Its values are

$$X_B(t_1) = 1, X_B(t_2) = 1, X_B(t_3) = 0, X_B(t_4) = 0, X_B(t_5) = 1, X_B(t_6) = 0.$$

A real-valued random variable is a measurable function from outcomes to real numbers, $X: \Omega \rightarrow \mathbb{R}$. We explain measurability below using inverse images. Its input is an individual outcome. In contrast, a **set function** has sets as inputs: the probability measure takes an event E and returns $P(E)$. Having a set as its domain does not make every function a set function. Once the outcome is specified, the function's value is fixed. It is not a random mechanism and it is not the probability measure. In this experiment, the ticket ID is selected at random; X_B records whether its reviewer reference includes Billing.

Image: values produced by a set of outcomes

For $E \subseteq \Omega$, the image of E under X_B is

$$X_B(E) = \{X_B(t): t \in E\}.$$

For the Account Support event $A = \{t_1, t_3, t_6\}$, t_1 also requires Billing, while t_3 and t_6 do not. Hence

$$X_B(A) = \{0, 1\}.$$

This is a set of numerical values, not an event in the ticket sample space.

Inverse image: outcomes that produce selected values

For $D \subseteq \mathbb{R}$, the inverse image is

$$X_B^{-1}(D) = \{t \in \Omega : X_B(t) \in D\}.$$

In particular,

$$X_B^{-1}(\{1\}) = \{t_1, t_2, t_5\} = B,$$

and

$$X_B^{-1}(\{0\}) = \{t_3, t_4, t_6\} = B^c.$$

An inverse image is a set of outcomes; it need not be an inverse function. The inverse image connects a numerical condition on a random variable to an event to which the probability measure applies:

$$P(X_B = 1) = P(X_B^{-1}(\{1\})) = P(B) = \frac{3}{6} = \frac{1}{2}.$$

The distribution of X_B is $P(X_B = 1) = 1/2$ and $P(X_B = 0) = 1/2$. The function X_B , its inverse-image event B , and the number $P(B)$ are three different objects.

Measurability and the probability law

To calculate $P(X_B \in D)$, the inverse image $X_B^{-1}(D)$ must be an event. For example, $X_B^{-1}((1/2, 3/2)) = B \in \mathcal{F}$, giving probability $1/2$. This requirement motivates the formal definition.

As in Section 5, now on $\mathbb{R}$, the **Borel σ -algebra** $\mathcal{B}(\mathbb{R})$ is the smallest sigma-algebra containing all open intervals. It includes intervals, singletons and their countable unions, intersections and complements. A function $X: \Omega \rightarrow \mathbb{R}$ is **measurable**, and hence a real-valued random variable on $(\Omega, \mathcal{F}, P)$, if

$$X^{-1}(D) \in \mathcal{F} \quad \text{for every } D \in \mathcal{B}(\mathbb{R}).$$

Its **law** or **distribution** is the probability measure

$$P_X(D) = P(X^{-1}(D)), \quad D \in \mathcal{B}(\mathbb{R}).$$

In the full finite model every function is measurable because all subsets are events. For X_B , the inverse image is one of $\emptyset, B, B^c, \Omega$ according to whether D contains neither of 0 and 1, only 1, only 0, or both. Thus X_B is measurable even for the smaller collection $\mathcal{G}$. Its law assigns mass $1/2$ each to 0 and 1. By contrast, the ID-code function $J(t_i) = i$ is not measurable for $\mathcal{G}$: $J^{-1}(\{1\}) = \{t_1\}$ is unavailable in that collection. It is measurable for 2^Ω .

Worked check: a die outcome and its numerical value

Roll a stipulated fair die once. Take $\Omega_d = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F}_d = 2^{\Omega_d}$ and $P_d(E) = |E|/6$. Define $Z(\omega) = \omega$. If face 4 occurs, distinguish the outcome 4, the singleton event $\{4\}$, the function value $Z(4) = 4$, and the probability $P_d(\{4\}) = 1/6$. Also $Z^{-1}(\{2, 4, 6\}) = \{2, 4, 6\}$, so $P_d(Z \text{ is even}) = 1/2$. The identical numerals in this identity function do not identify its values with their probabilities.

Checkpoint 5. Find $X_B(\Omega)$, $X_B(A)$, $X_B^{-1}(\{1\})$ and $P(X_B = 1)$. Identify the sets of real values, the set of ticket outcomes, and the probability.

7. A continuous example: ticket closure time

The triplet is often not written out in routine probability calculations. To see what remains implicit, consider a ticket selected from a specified service process, and let T_c be its closure time, measured in hours. The question

What is the probability that this ticket takes more than one day to close?

is written $P(T_c > 24)$, since one day is 24 hours. In an underlying ticket-system experiment, each outcome ω determines a closure time $T_c(\omega)$ in hours. The event of interest is the inverse image

$$T_c^{-1}((24, \infty)) = \{\omega \in \Omega: T_c(\omega) > 24\}.$$

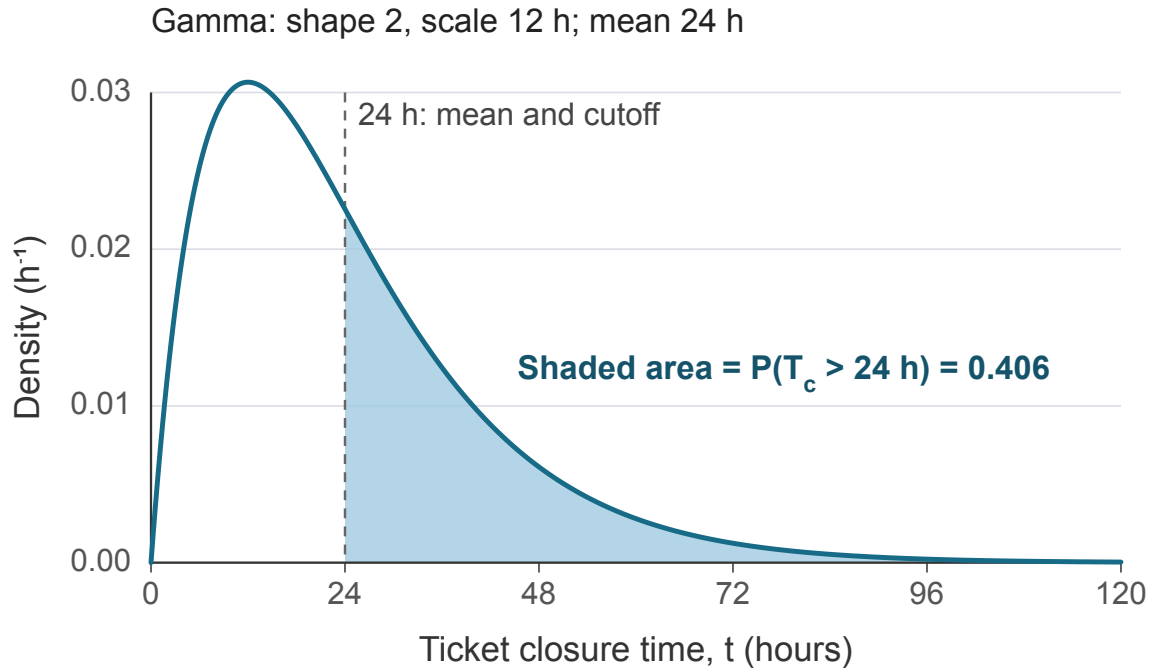
For an illustrative calculation, suppose the closure time follows a Gamma distribution with shape $\alpha = 2$ and scale $\theta = 12$ hours. These are hypothetical model parameters, not estimates from the six-ticket audit. Under the shape-scale parameterization, the density is

$$f_{T_c}(t) = \frac{t^{\alpha-1} e^{-t/\theta}}{\Gamma(\alpha) \theta^\alpha}, \quad t \geq 0.$$

Here $\alpha, \theta > 0$ and $\Gamma(a) = \int_0^\infty u^{a-1} e^{-u} du$ for $a > 0$; in particular, $\Gamma(2) = 1$. With $\alpha = 2$ and $\theta = 12$ hours, this becomes

$$f_{T_c}(t) = \frac{t}{12^2} e^{-t/12}, \quad t \geq 0,$$

where t is measured in hours. Here α is the shape parameter, not the mean. The mean in this shape-scale parameterization is $E[T_c] = \alpha\theta = 24$ hours. The plotted curve shows this model and shades the area to the right of the 24-hour cutoff, which coincides with the mean.



Gamma density for ticket closure time T_c under the illustrative model. The vertical line marks both the mean and the 24-hour cutoff; the shaded right tail represents $P(T_c > 24 \text{ h})$.

For the six-ticket experiment above, probabilities were calculated by counting equally likely IDs: $P(C) = |C|/6$. A continuous ticket time cannot be handled by counting individual possible durations. Under a density model, the probability of an interval is the area under the density over that interval. A density value is not itself a probability; integration accumulates the density across the range. In the audit model, the corresponding finite calculation is a sum of singleton probabilities; for example, $P(C) = \sum_{t_i \in C} P(\{t_i\}) = 4/6$. The closure-time probability is the integral of its density over the event. With the substitution $u = t/12$,

$$\begin{aligned} P(T_c > 24) &= \int_{24}^{\infty} \frac{t}{144} e^{-t/12} dt \\ &= \int_2^{\infty} u e^{-u} du \\ &= \left[-(u+1)e^{-u} \right]_2^{\infty} \\ &= (2+1)e^{-2} = 3e^{-2} \approx 0.406. \end{aligned}$$

Thus, under this stipulated model, the probability that closure takes more than 24 hours is about 0.406. A different shape, scale or target ticket population would give a different value. Specifying the Gamma family alone would not suffice; the parameters and their parameterization are needed.

What a probability density specifies

The calculation above uses an area, not a density height. Formally, a random variable T has a **probability density function** f if f is nonnegative and measurable, $\int_{\mathbb{R}} f(t) dt = 1$, and

$$P(T \in D) = \int_D f(t) dt \quad \text{for every Borel set } D.$$

Here $f(t) = 0$ for $t < 0$. The same substitution checks normalization: $\int_0^\infty (t/144)e^{-t/12} dt = \int_0^\infty ue^{-u} du = 1$. A singleton has zero area, so $P(T_c = 24) = 0$, while a range can have positive probability. Not every random variable has a density; the Billing indicator has a discrete law.

The triplet for the closure-time model

For this calculation, use the following canonical probability space, which models closure time alone:

$$(\Omega_c, \mathcal{F}_c, P_c).$$

In the calculation above, the shorthand $P(T_c > 24)$ denotes $P_c(T_c > 24)$ on this space. It is distinct from P , the uniform measure used for selecting one of the six audited ticket IDs.

1. **Sample space:** $\Omega_c = [0, \infty)$, the possible closure durations measured in hours.
2. **Event collection:** $\mathcal{F}_c = \mathcal{B}([0, \infty))$, the Borel subsets of the nonnegative real line: sets obtained by intersecting Borel subsets of $\mathbb{R}$ with $[0, \infty)$.
3. **Probability measure:** for every $E \in \mathcal{F}_c$,

$$P_c(E) = \int_E f_{T_c}(t) dt, \quad f_{T_c}(t) = \frac{t}{144} e^{-t/12}, \quad t \geq 0.$$

The density integrates to one, so $P_c(\Omega_c) = 1$. Define the random variable by $T_c: \Omega_c \rightarrow [0, \infty)$, $T_c(t) = t$. Thus the event of closure after 24 hours is

$$T_c^{-1}((24, \infty)) = (24, \infty) \in \mathcal{F}_c,$$

and its probability is

$$P_c(T_c > 24) = P_c((24, \infty)) = \int_{24}^\infty f_{T_c}(t) dt.$$

This is one explicit probability space realizing the assumed Gamma law. A full ticket-system model could use outcomes that also include ticket type, priority and other information, with closure time as a function on that larger space. In routine calculations, $T_c \sim \text{Gamma}(\text{shape } 2, \text{scale } 12 \text{ h})$ states the induced law needed for probabilities about closure time without writing the whole triplet. The distribution of T_c does not by itself describe every other random quantity in the ticket system.

This is why many engineering and AI calculations work directly with random variables and their distributions after the probability-space foundation has been established.

8. Worked problems

Problem 1: probability of "Other"

Find the probability that a uniformly selected ticket has neither Billing nor Account Support required according to the reviewer.

Solution. By definition, $O = (B \cup A)^c = \{t_4\}$. Therefore

$$P(O) = \frac{1}{6}.$$

Problem 2: exactly one named destination is required

Find the event and probability that exactly one of Billing and Account Support is required.

Solution. The event is

$$(B \setminus A) \cup (A \setminus B) = \{t_2, t_3, t_5, t_6\}.$$

Its probability is $4/6 = 2/3$.

Problem 3: agreement between reviewer and model

Let $M(t)$ be the model's classification and $R(t)$ the reviewer's reference classification for ticket t . Define exact agreement as equality of these classification sets. The agreement event from the table is

$$C = \{t_2, t_3, t_5, t_6\}.$$

Under uniform selection, $P(C) = 4/6 = 2/3$. This equals the observed agreement fraction because the experiment selects uniformly from this exact audit.

Problem 4: use an inverse image

Define $Y(t) = 1$ if the model classification exactly matches the reviewer and $Y(t) = 0$ otherwise. Then $Y^{-1}(\{1\}) = C$, and

$$P(Y = 1) = P(C) = \frac{4}{6} = \frac{2}{3}.$$

The random variable records an outcome numerically; its inverse image gives the event; the probability measure assigns that event a number.

Problem 5: change the audit selection weights

In a hypothetical follow-up experiment, select the same six IDs with stipulated weights $(1, 1, 1, 3, 1, 1)/8$. Keep Ω , its full power set and the labels fixed. Find the probabilities of agreement and Other.

Solution. The weights are nonnegative and sum to one, defining $Q(E) = \sum_{t_i \in E} w_i$. Therefore $Q(C) = (1 + 1 + 1 + 1)/8 = 1/2$ and $Q(O) = 3/8$. The observed fraction remains $4/6$, but the selection probability changes. These weights specify an exercise, not an estimate of future traffic.

Problem 6: fraud-screen flags

A hypothetical fixed review batch has 20 transaction IDs. Reviewers label 4 as fraudulent. A screen flags 5: 3 of the 4 reviewer-positive transactions and 2 reviewer-negative transactions. Select one ID uniformly. Find the false-positive and false-negative events' probabilities.

Solution. On $\Omega = \{1, \dots, 20\}$ with its power set and uniform measure, write F for flagged IDs and H for reviewer-positive IDs. Then $|F \cap H^c| = 2$ and $|F^c \cap H| = 1$, so their probabilities are $2/20 = 0.10$ and $1/20 = 0.05$. Among the 16 reviewer-negative records, the false-positive fraction is instead $2/16 = 0.125$. That changes the denominator and the selection experiment; it is not the probability just calculated.

Problem 7: sensor tolerance

For a hypothetical engineering model, a sensor error ε is uniform on $[-2, 2]$ millimetres. Use this interval as the sample space, its Borel sets as events, and density $f(e) = 1/4$ on the interval. What is the probability of exceeding a tolerance of 1 mm in absolute value?

Solution. The error is the identity random variable. The failure event is $[-2, -1) \cup (1, 2]$, of total length 2, giving $P(|\varepsilon| > 1) = \int_{[-2, -1) \cup (1, 2]} (1/4)de = 1/2$. Exact boundary values have probability zero under this model. Uniform error is stipulated, not established by the tolerance specification.

Problem 8: assessment pass indicator

Select one of five hypothetical assessment records uniformly. Their fixed scores are 45, 60, 70, 80 and 95; the pass threshold is 60 inclusive. Specify a random variable and its law for the pass/fail result.

Solution. Outcomes are record IDs $a_1, \dots, a_5$, with the power set and singleton weights $1/5$. Set $Y(a_i) = \mathbf{1}_{\{\text{score}(a_i) \geq 60\}}$. Then $Y^{-1}(\{1\}) = \{a_2, a_3, a_4, a_5\}$, $P_Y(\{1\}) = 4/5$ and $P_Y(\{0\}) = 1/5$. This is a probability under selection from these records; it does not establish a future cohort's pass rate.

Practice with equally likely outcomes

For each exercise below, first specify the elementary outcomes and the sampling assumptions. Use the finite triplet

$$(\Omega, 2^\Omega, P), \quad P(\{\omega\}) = \frac{1}{|\Omega|}, \quad P(E) = \frac{|E|}{|\Omega|}.$$

The stated mechanisms make the elementary toss sequences, ordered die pairs, individual cards, ordered draws or selected subsets equiprobable. Records and summaries computed from those outcomes need not be equally likely. Fairness and independence are stipulated for coins and dice; uniform selection is stipulated for cards and people.

For successive uniform draws of two distinct objects from n objects, use the rule that each ordered pair has probability $1/[n(n-1)]$. We stipulate this rule here; its product form $(1/n)(1/(n-1))$ will be justified using conditional probability. These practice questions have no worked answers in the student materials.

When a selection has no order, $\binom{n}{k}$ denotes the number of k -element subsets of n distinct objects. In particular, $\binom{n}{2} = n(n-1)/2$; justify the factor of 2 when comparing ordered roles with committees in P3.

Coins

C1. Two tosses and overlapping events. Toss a fair coin twice independently. Let E be "the first toss is heads" and F be "exactly one head." List $\Omega, E, F, E \cap F$ and $E \cup F$, and calculate their event probabilities. Are E and F mutually exclusive?

C2. Three tosses. Toss a fair coin three times independently. Find the probabilities of exactly two heads, at least two heads, and at least one head. Calculate the last probability both by counting directly and by using the complement.

C3. From sequences to a random variable. For the same three-toss experiment, define $N(\omega)$ as the number of heads. List $N^{-1}(\{k\})$ for $k = 0, 1, 2, 3$ and give the law of N . Explain why assigning probability $1/4$ to each possible head count would describe a different model.

Dice

D1. One die. Roll one stipulated fair six-sided die. Let E be "an even face" and F be "a face greater than 3." List their intersection and union. Find $P(E \cup F)$ and the probability of exactly one of E, F occurring.

D2. Two distinguishable dice. Roll independent fair red and blue six-sided dice. Use ordered pairs (r, b) as outcomes. Find the probabilities of a total of 7, a double, and a total of at least 10. Show the favorable pairs or their counts.

D3. Forgetting the colours. Keep D2's experiment, but record $S(r, b) = (\min(r, b), \max(r, b))$, the two faces in increasing order, allowing equality. How many possible records are there? Are these records equally likely? Find $P(S = (1, 1))$ and $P(S = (1, 2))$ by counting their preimages in the original 36-outcome space, then calculate the probability of recording at least one 1.

A deck of cards

Use a standard deck of 52 distinct cards with no jokers: 13 ranks in each of four suits, with one ace per suit. In each draw, choose uniformly from the cards currently available. In K2 and K3 the two cards are drawn **without replacement**.

K1. One card and a union. Specify the sample space using card identities. Find the probability that the card is an ace or a heart. Explain why adding $4/52$ and $13/52$ without correction is wrong.

K2. Two ordered draws. Record the first and second cards separately. How many elementary outcomes are there, and why are they equally likely? Find the probabilities that both cards are aces and that the first is an ace while the second is not.

K3. An unordered two-card hand. Select a two-card subset uniformly. Find the probability of exactly one ace. Recalculate it using ordered draws without replacement and verify agreement. Explain why “zero aces”, “one ace” and “two aces” are not three equiprobable outcomes.

Selecting people

In a hypothetical training group, six distinct participants have IDs a, b, c, d, e, f . Four, a, b, c, d , have completed a specified training module; e, f have not. These are fixed labels. Every selection below follows the stated uniform mechanism.

P1. One participant. Select one of the six IDs uniformly. Find the probability that the selected participant has completed the module and the probability that the selected ID is a or f . Distinguish an individual outcome from the event “completed.”

P2. A two-person committee. Select a two-element subset uniformly from the six participants. Find the probabilities that both have completed the module, exactly one has completed it, and neither has completed it. Check that these probabilities sum to 1. Why should the denominator count committees rather than completion categories?

P3. Two distinct roles. Select a chair uniformly from all six participants, then a secretary uniformly from the remaining five. Use ordered pairs of distinct IDs. Find the probabilities that the chair has completed the module and the secretary has not, and that the two office holders are a and b in either order. If the role labels are discarded, is the resulting committee uniform over the 15 pairs? Justify your answer by counting preimages.

Reflection

1. Which assumptions establish equiprobability in each experiment?
2. Does uniform sampling of elementary outcomes make head counts, dice totals or committee compositions uniform? Explain using an exercise above.
3. When does discarding order preserve uniformity? Compare dice records, two-card hands and two-person committees.
4. A hypothetical evaluation set has 8 cases of one task type and 2 of another. Under uniform selection of a case ID, are the two task types equally likely? Instead choose either task type with probability $1/2$, then select a case uniformly within that type. Find the probabilities of each type and of each individual case under this second rule.

9. Connection to AI evaluation and continuation

The probability $2/3$ here concerns uniform selection from six audited ticket IDs. It is not automatically the model’s agreement probability on future traffic. That requires a defined target population, an appropriate sampling process and assumptions connecting this audit to future tickets. Changing the reviewer protocol may also change the event called “agreement.”

The same distinctions arise in AI evaluation. A test-set accuracy is an observed proportion. Interpreting it as a probability of correct classification for future cases requires additional assumptions. If model generation is stochastic, its repeated-output distribution is another probability model with a different outcome space. In each case, specify the outcomes, event and sampling or generation process before interpreting a probability.

A suitable first stopping point is the calculation of $P(B \cup A) = 5/6$ at the end of Section 4. Continue to the triplet, sigma-algebra and axioms once the class can identify the complement event O . The random-variable and image sections can continue in a later meeting. Bertrand’s chord problem is posed here but not derived; the rigorous supplement specifies its sampling models and works through their parametrizations and probabilities.

References

- Jay L. Devore, *Probability and Statistics for Engineering and the Sciences*. 8th edition: Chapter 2, "Probability"; §3.1, "Random Variables"; §4.1, "Probability Density Functions."
- Krishna Jagannathan, *Probability Foundations for Electrical Engineers*, lecture notes, 9 July 2025: Chapter 1 (sets and functions), Chapter 4 (probability models), Chapter 5 (probability rules), Chapter 6 (discrete models), and Chapter 11, Definitions 68–70 (measurable functions and laws).
- Alan Hájek, "Interpretations of Probability," *Stanford Encyclopedia of Philosophy*, comparative overview: <https://plato.stanford.edu/entries/probability-interpret/>.
- John D. Norton, "Bertrand's Geometrical Paradoxes: Proofs," University of Pittsburgh, presenting the chord constructions yielding $1/3$, $1/2$ and $1/4$: https://sites.pitt.edu/~jdnorton/teaching/paradox/chapters/probability_for_indifference/Bertrand_proof/Bertrand_proof.html.

The supplement gives proofs and further boundary cases. All audit data, model parameters and practice records in this lesson are hypothetical and stipulated for teaching.